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UNIVERSITY OF BATH, ARCHITECTURE AND CIVIL ENGINEERING

AR10368 Surveying

Deformation Monitoring Coursework

Building Deformation Monitoring

You are responsible for monitoring the condition of the Holburne Museum during construction of a Bath Rapid Transit railway, linking Bath Spa University College to the University of Bath via the city centre. The tunnel will run directly under the centreline of the Museum from west to east (Figure 1).

Tunnelling under the Museum is expected to take 2 days in total. Clearly there is a need to ensure distortion of the structure due to tunnelling activity will not result in damage to the structure. You need to design a deformation monitoring plan to detect any deformation of the Museum that could lead to long term damage and halt the works. Differential settlement greater than 10 mm is a particular concern.

You might wish to consider other potential sources of deformation to ensure you don't give a false alarm. For example, a structural analysis has indicated that the maximum likely horizontal deflections of the Museum due to wind loading and thermal expansion are 2.5 mm and 6.5 mm respectively.

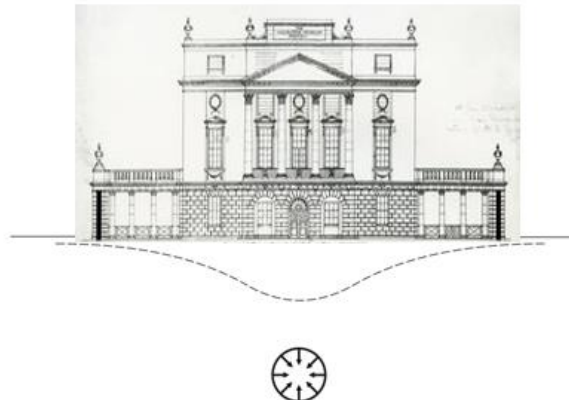


Figure 1. Tunnel running west to east under the centreline of the Holburne Museum.

Provide a plan for monitoring the condition of the Holburne Museum as a result of the tunnelling works. Include the methods used and why, as well as details of a suggested measurement programme. You may use figures if they are helpful. Use appropriate sub-headings to structure your answer.

ANSWER

Text must be provided in the space below;

Research of structure:

To understand the deformations that are likely to occur we must analyse the building such that we can predict where they are likely to propagate and by how much. We can see how the earth beneath the museum is likely to move in response to the tunnelling in Figure 1. This shows that the deformation is going to be non-uniform with much larger settlements to be experienced towards the centre of the building. This type of differential settlement will mean that thin vertical elements such as the columns will mostly just move vertically relative to one another. Therefore, they will not be the elements that are likely to crack, instead the cracks will propagate in the horizontal elements that connect adjacent columns as they will have relative movement from one end to the other. This can be seen in figure 2. However wider vertical sections such as those between windows will be large enough that the shear force due to the settlement can cause cracking such as that in Figure 3. These same principles can be applied to the side faces of the building, as tunnelling is underway there will be differential settlement as the boring machine moves from under one side of the building to the other, as shown in figure 4. Leading to similar cracking as the front face. We must also understand that not all deformation will be due to the tunnelling, some deformation will be due to thermal expansion, wind loading, people moving about in the building, possible localised plant growth and other asymmetrical loading, an exaggerated example of which can be seen in figure 5.

Technique:

As the tunnelling is over a relatively short time period, 2 days, the deformation caused is also likely to occur rapidly. Therefore, we need a surveying procedure that is only over a short time period but will have many survey points and many repetitions of measurements. This is so that any movement can be noticed before the deformation becomes problematic. In order to do this, I decided that the use of total stations, dumpy levels and extensometers would be most appropriate. These instruments are more accurate than necessary to notice the maximum allowable deformation of 10mm (theodolite with an accuracy of 1.5 millimetres plus/minus 2 parts per million₁, level with an accuracy of 1:4000 for every 100 meters₂, extensometers with resolutions of 0.3µm₃). They are also far cheaper than more advanced systems and they are quick and easy to set up and operate, therefore not needing as many skilled workers on site. Also, as we are only surveying one building it would not be a viable option to use equipment such as lidar and GPS, as the price is only justifiable when using them on a wider scale.

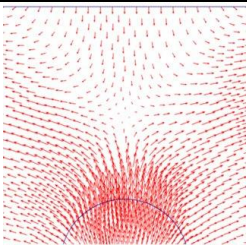
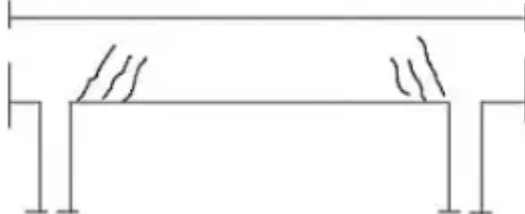
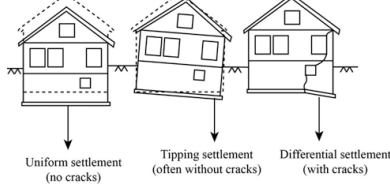
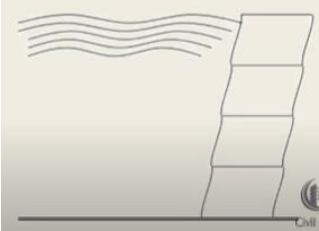
We would also use temperature and wind gauges, so that we can see how much deformation is likely to occur due to these forces and make necessary adjustments to our calculations.

Plan of measurement points:

As deformation is likely to occur on all faces of the building, we must be able to monitor them all and therefore due to obstructions, a total station will have to be placed in the 4 positions shown in figure 6. The total stations in positions A and C will also have to have their levels checked by dumpy level, from E and F, and G and H respectively. This is because although they have been moved from the likely line of deformation due to tunnelling, there may still be some settlement. Therefore any change in height of the total station must be noted as this will affect the reading that they output. The points that will be surveyed using the total station will be those at the point where vertical and horizontal elements meet as this is where relative movement is most important, such as edges of windows and tops of columns. These points will have a reflective prism mounted on them, such that the total station can work out the distances and angles automatically using its built-in scanner. Finally, there will be several extensometers that will be placed in areas of existing cracks, and where cracks are likely to occur as shown in figure 2 before.

Surveying schedule:

The first set of readings, one at mid-day or at high temperature and one in the evening or at low temperature, will be taken before the tunnelling has commenced in order to give us a base set of values. These values will be before any deformation due to tunnelling has occurred, therefore showing if our research of how the building will deform due to other factors is correct. During tunnelling there will be 5 sets of readings on both days, giving enough data and time to be able to stop excavation if the deformations are reaching the allowable limits, without having to use more complex and expensive continual data analysis. There will also be one additional survey each day for 3 days after tunnelling, as this will allow for any smaller amounts of settlement to be noticed and the ground should not settle any more after these 3 days.

	Figure 1 – Excessive grout pressures around the lining of shield driven tunnels in soft soils. [online] Available from: https://www.researchgate.net/figure/Vector-plot-of-the-soil-deformation-above-the-tunnel_fig4_299809190 [accessed Apr, 2022]
	Figure 2 – TheConstructor(2017).Types of Cracks in Concrete Beams and their Causes [online] Available from: https://theconstructor.org/concrete/types-of-cracks-in-concrete-beams/5948/ [accessed Apr, 2022]
	Figure 3 - Anon, (n.d.). <i>The Foundation Works - Providing Effective Foundation Solutions</i>. [online] Available at: https://www.thefoundationworks.com/common-issues/cracking-walls/. [accessed Apr, 2022]
	Figure 4 – University of bath lecture https://moodle.bath.ac.uk/pluginfile.php/2023956/mod_resource/content/0/Deformation%20and%20Beach%20Monitoring%20In%20Person_Final.pdf
	Figure 5 – www.youtube.com. (n.d.). Lecture 1 Dynamic Loads, Earthquake & Plate Tectonics [Structural Mechanics]. [online] Available at: https://www.youtube.com/watch?v=iAwbw_nKero [Accessed Apr. 2022].
	Figure 6 – Edited google maps image https://www.google.com/maps/place/The+Holburne+Museum/@51.3858137,-2.3512003,115m/data=!3m1!1e3!4m5!3m4!1s0x4871810cc0d79c9f:0x71fe65e7d6e24ce0!8m2!3d51.3857024!4d-2.3509979

References:

- 1 - Elhassan, I.M. and Ali, A.S. (2011). Comparative study of accuracy in distance measurement using: Optical and digital levels. *Journal of King Saud University - Engineering Sciences*, [online] 23(1), pp.15–19. Available at: <https://www.sciencedirect.com/science/article/pii/S1018363910000048> [Accessed . 2019].
- 2 - Civil Engineering Portal - Biggest Civil Engineering Information Sharing Website. (2022). *Dumpy Level*. [online] Available at: <https://www.engineeringcivil.com/dumpy-level.html#:~:text=The%20primary%20advantage%20of%20using> [Accessed Apr. 2022].
- 3 - *A Users Guide to Video Extensometers*. [online] Available at: <https://www.ametektest.com/learningzone/library/articles/a-users-guide-to-video-extensometers> [Accessed Apr. 2022].